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Research Interests

- My research focuses on poroelasticity and myocardial perfusion modeling, emphasizing coronary circulation and its coupling with systemic hemodynamics. I develop computational models to study the mechanics and transport processes in perfused tissues and aim to translate these high-fidelity simulations into clinically useful tools for diagnosis, treatment planning, and personalized medicine.

Employment History

2026 – ... 📌 Actively seeking employment opportunities.

Education

- 2022 – 2026 📌 **Ph.D.** in Mathematics.
School of Mathematics and Statistics, Northwestern Polytechnical University
- 2019 – 2022 📌 **M.Sc.** in Applied Mathematics
School of Mathematics, Northwestern University
- 2015 – 2019 📌 **B.Sc.** in Mathematics and Applied Mathematics
School of Mathematics, Northwestern University

Research Publications

Journal Articles

- 1 P. Ma, L. Cai, **X. Wang**, and H. Gao, "AFSI: Automated fluid-structure interaction solver development for nonlinear solid mechanics," *arXiv preprint arXiv:2509.00014*, 2025.
- 2 P. Ma, L. Cai, **X. Wang**, and H. Gao, "Fully GPU-accelerated, matrix-free immersed boundary method for complex fiber-reinforced hyperelastic cardiac models," *Computer Methods in Applied Mechanics and Engineering*, vol. 447, p. 118 353, 2025, ISSN: 0045-7825.
- 3 **X. Wang**, L. Cai, P. Ma, and H. Gao, "An efficient staggered scheme for solving the poromechanics problem of quasi-static cardiac perfusion," *International Journal for Numerical Methods in Biomedical Engineering*, vol. 41, no. 4, e70030, 2025.
- 4 P. Ma, L. Cai, **X. Wang**, Y. Wang, X. Luo, and H. Gao, "An unconditionally stable scheme for the immersed boundary method with application in cardiac mechanics," *Physics of Fluids*, vol. 36, no. 8, 2024.

Skills

- Coding 📌 C++, Python, MATLAB, $\text{\LaTeX}$
- Numerical 📌 Finite Element Method, Finite Difference Method, Immersed Boundary Method

Miscellaneous Experience

Participated Projects

- 2019.1–2022.12  National Natural Science Foundation of China General Program (11871399), Efficient Numerical Methods and Applications for Electrophysiological Coupling in Injured Hearts
- 2023.1–2026.12  National Natural Science Foundation of China General Program (12271440), Multiscale and Multiphysics Coupled Modeling and Computation of Myocardial Perfusion

Software Development

- AFSI  An open-source Python package for automated fluid-structure interaction modeling.
<https://github.com/npuheart/afsi>
- NPUHEART  A high-performance GPU-based framework for large-scale fluid-structure interaction simulations.
(Not yet open-sourced)

References

- Supervisors  <https://teacher.nwpu.edu.cn/caili.html>
- Publication  <https://scholar.google.com.hk/citations?user=x82NMcwAAAAJ>
- Collaborators  <https://www.maths.gla.ac.uk/~nah/>
<https://haogao.github.io>